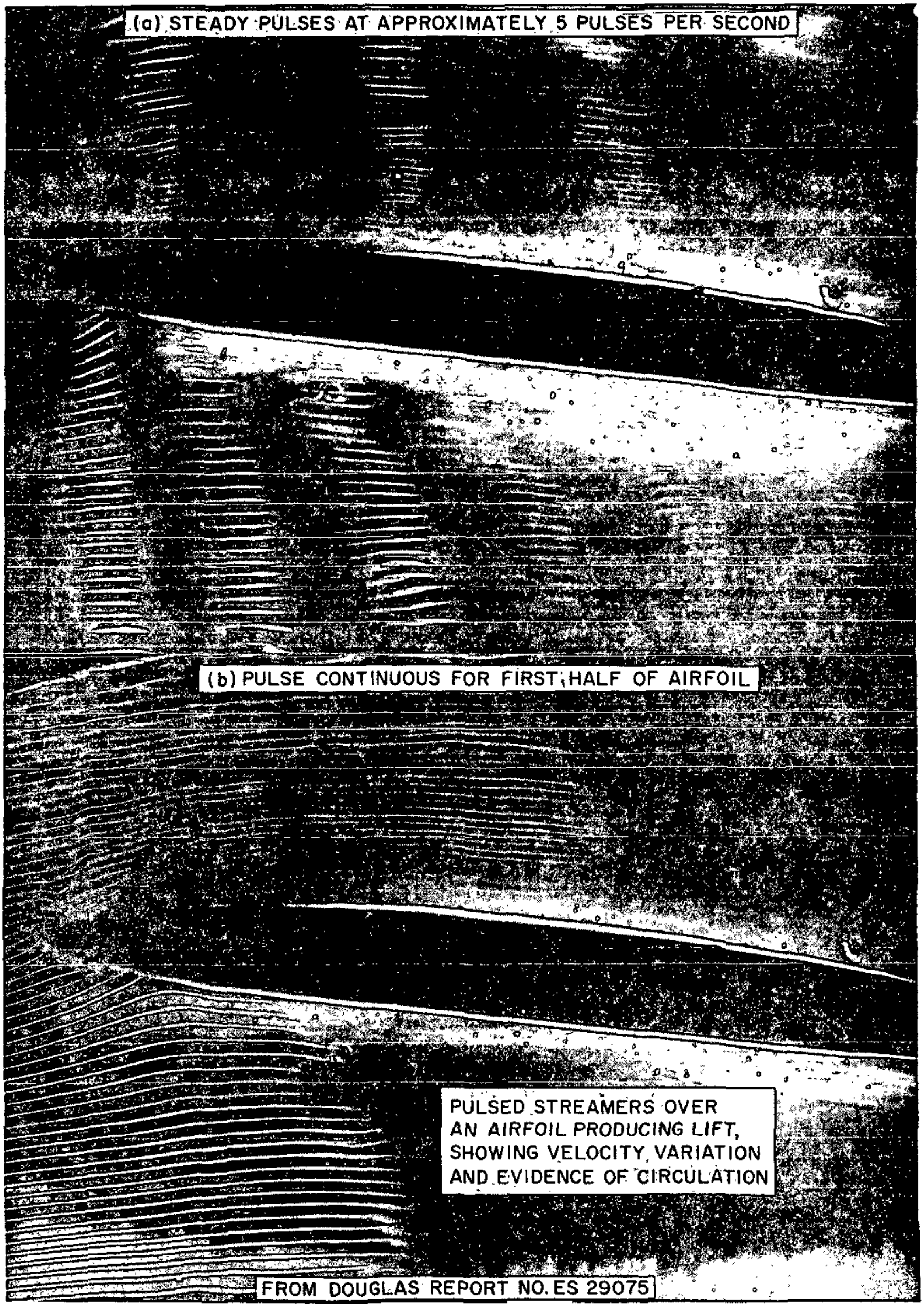


Figure 1.8. Generation of Lift (sheet 2 of 2)

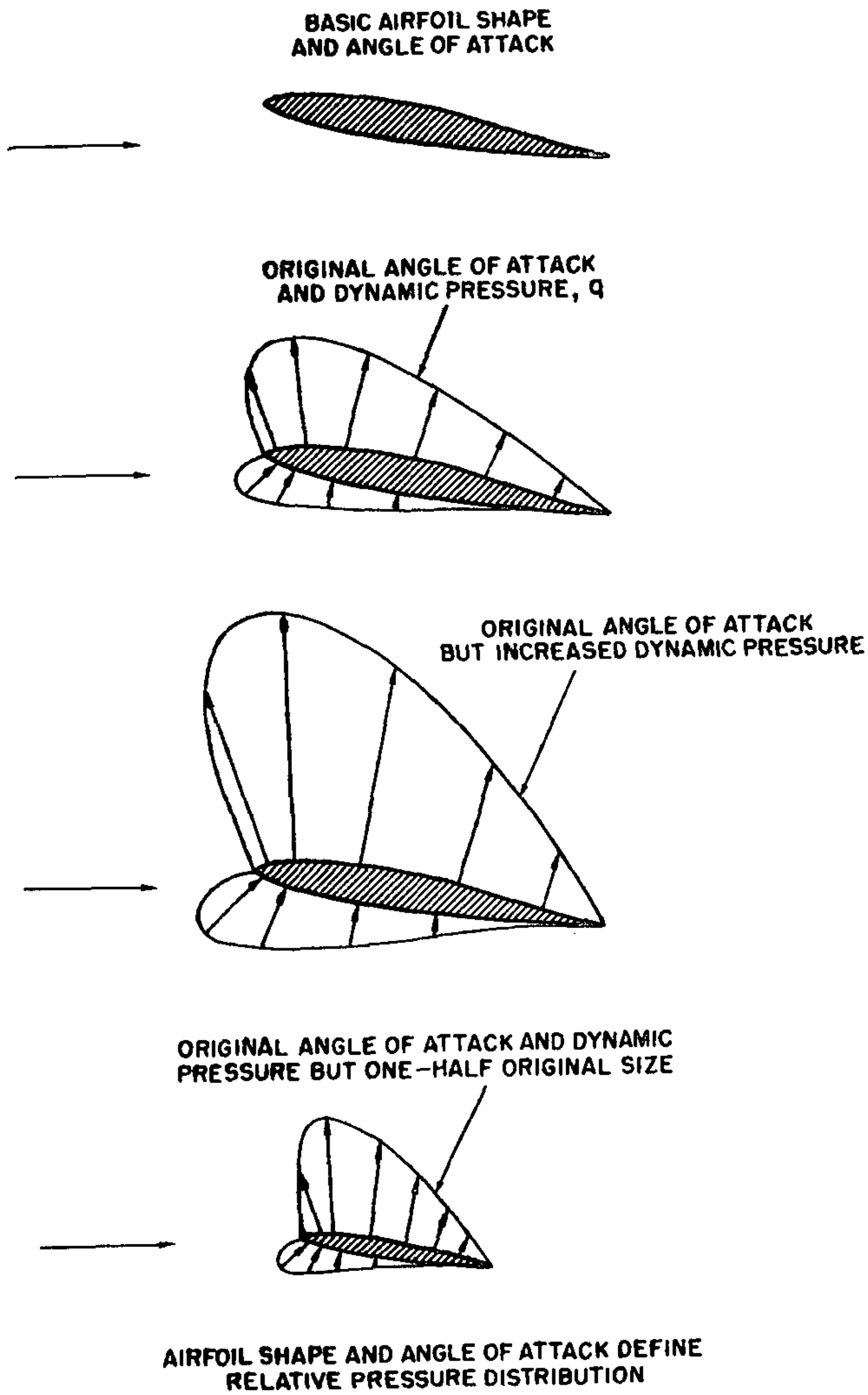


Figure 1.9. Airfoil Pressure Distribution

rotation will be quite a "curve ball artist" the golfer that cannot control the lateral motion of the club face striking the golf ball will impart an uncontrollable spin and have trouble with a "hook" or "slice."

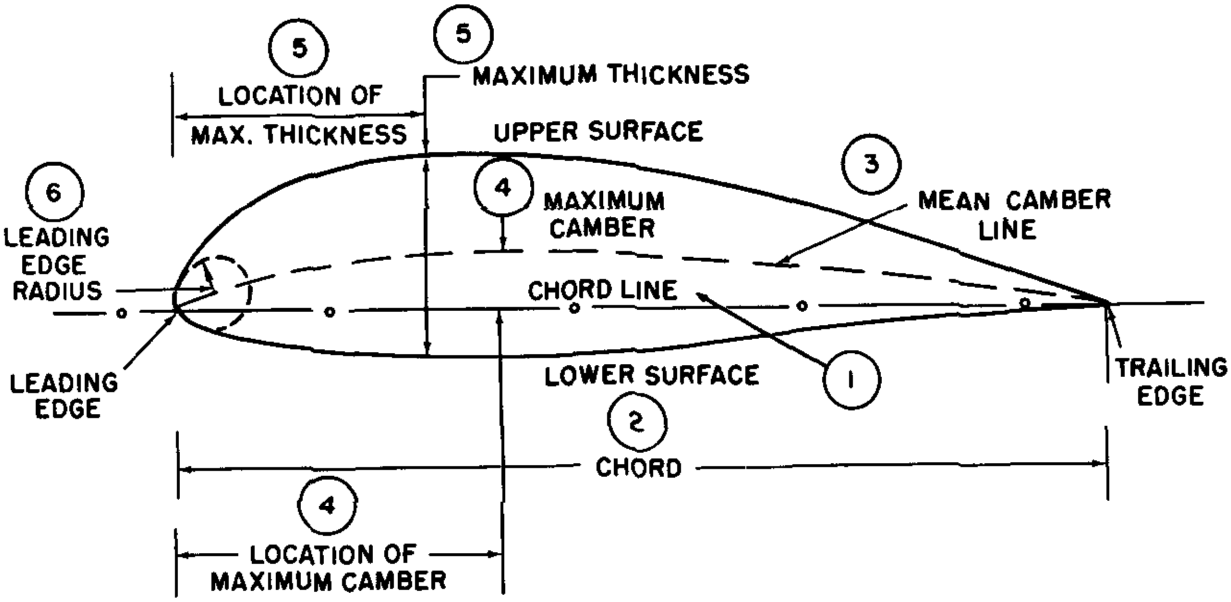
While a rotating cylinder can produce a net lift from the circulatory flow, the method is relatively inefficient and only serves to point out the relationship between lift and circulation. An airfoil is capable of producing lift with relatively high efficiency and the process is illustrated in figure 1.8. If a symmetrical airfoil is placed at zero angle of attack to the airstream, the streamline pattern and pressure distribution give evidence of zero lift. However, if the airfoil is given a positive angle of attack, changes occur in the streamline pattern and pressure distribution similar to changes caused by the addition of circulation to the cylinder. The positive angle of attack causes increased velocity on the upper surface with an increase in upper surface suction while the decreased velocity on the lower surface causes a decrease in lower surface suction. Also, upwash is generated ahead of the airfoil, the forward stagnation point moves under the leading edge, and a downwash is evident aft of the airfoil. The pressure distribution on the airfoil now provides a net force perpendicular to the airstream—lift.

The generation of lift by an airfoil is dependent upon the airfoil being able to create circulation in the airstream and develop the lifting pressure distribution on the surface. In all cases, the generated lift will be the net force caused by the distribution of pressure over the upper and lower surfaces of the airfoil. At low angles of attack, suction pressures usually will exist on both upper and lower surfaces but the upper surface suction must be greater for positive lift. At high angles of attack near that for maximum lift, a positive pressure will exist on the lower surface but this will account for approximately one-third the net lift.

The effect of free stream density and velocity is a necessary consideration when studying the development of the various aerodynamic forces. Suppose that a particular shape of airfoil is fixed at a particular angle to the airstream. The *relative* velocity and pressure distribution will be determined by the shape of the airfoil and the angle to the airstream. The effect of varying the airfoil size, air density and airspeed is shown in figure 1.9. If the same airfoil shape is placed at the same angle to an airstream with twice as great a dynamic pressure the *magnitude* of the pressure distribution will be twice as great but the *relative* shape of the pressure distribution will be the same. With twice as great a pressure existing over the surface, all aerodynamic forces and moments will double. If a half-size airfoil is placed at the same angle to the original airstream, the magnitude of the pressure distribution is the same as the original airfoil and again the *relative* shape of the pressure distribution is identical. The same pressure acting on the half-size surface would reduce all aerodynamic forces to one-half that of the original. This similarity of flow patterns means that the stagnation point occurs at the same place, the peak suction pressure occurs at the same place, and the actual *magnitude* of the aerodynamic forces and moments depends upon the airstream dynamic pressure and the surface area. This concept is extremely important when attempting to separate and analyze the most important factors affecting the development of aerodynamic forces.

AIRFOIL TERMINOLOGY. Since the shape of an airfoil and the inclination to the airstream are so important in determining the pressure distribution, it is necessary to properly define the airfoil terminology. Figure 1.10 shows a typical airfoil and illustrates the various items of airfoil terminology

(1) The *chord line* is a straight line connecting the leading and trailing edges of the airfoil.



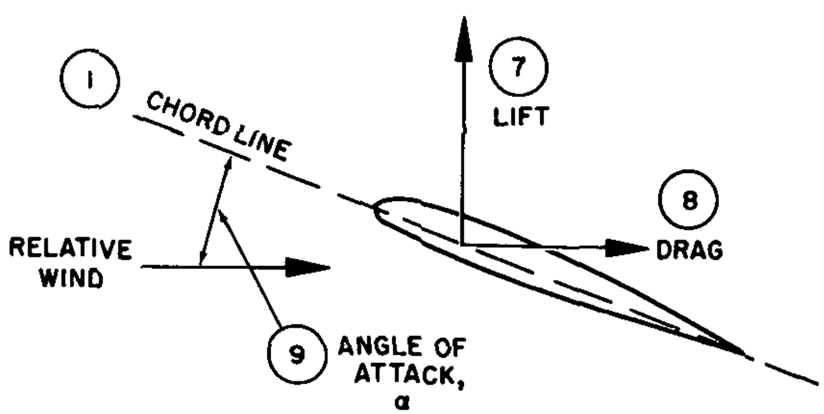


Figure 1.10. Airfoil Terminology

(2) The *chord* is the characteristic dimension of the airfoil.

(3) The *mean-camber line* is a line drawn halfway between the upper and lower surfaces. Actually, the chord line connects the ends of the mean-camber line.

(4) The shape of the mean-camber line is very important in determining the aerodynamic characteristics of an airfoil section. The *maximum camber* (displacement of the mean line from the chord line) and the *location* of the maximum camber help to define the shape of the mean-camber line. These quantities are expressed as fractions or percent of the basic chord dimension. A typical low speed airfoil may have a maximum camber of 4 percent located 40 percent aft of the leading edge.

(5) The thickness and thickness distribution of the profile are important properties of a section. The *maximum thickness* and *location* of maximum thickness define thickness and distribution of thickness and are expressed as fractions or percent of the chord. A typical low speed airfoil may have a maximum thickness of 12 percent located 30 percent aft of the leading edge.

(6) The *leading edge radius* of the airfoil is the radius of curvature given the leading edge shape. It is the radius of the circle centered on a line tangent to the leading edge camber and connecting tangency points of upper and lower surfaces with the leading edge. Typical leading edge radii are zero (knife edge) to 1 or 2 percent.

(7) The *lift* produced by an airfoil is the net force produced *perpendicular* to the *relative wind*.

(8) The *drag* incurred by an airfoil is the net force produced *parallel* to the *relative wind*.

(9) The *angle of attack* is the angle between the chord line and the relative wind. Angle of attack is given the shorthand notation α (alpha). Of course, it is important to differentiate between pitch attitude angle and angle of attack. Regardless of the condition of flight, the instantaneous flight path of the surface determines the direction of the oncoming relative wind and the angle of attack is the angle between the instantaneous relative wind and the chord line. To respect the definition of angle of attack, visualize the flight path of the aircraft during a loop and appreciate that the relative wind is defined by the flight path at any point during the maneuver.

Notice that the description of an airfoil profile is by dimensions which are fractions or percent of the basic chord dimension. Thus, when an airfoil profile is specified a *relative* shape is described. (NOTE: A numerical system of designating airfoil profiles originated by the National Advisory Committee for Aeronautics [NACA] is used to describe the main geometric features and certain aerodynamic properties. NACA Report No. 824 will provide the detail of this system.)

AERODYNAMIC FORCE COEFFICIENT. The aerodynamic forces of lift and drag depend on the combined effect of many different variables. The important single variables could be:

(1) Airstream velocity
(2) Air density
(3) Shape or profile of the surface
(4) Angle of attack
(5) Surface area
(6) Compressibility effects
(7) Viscosity effects

If the effects of viscosity and compressibility are not of immediate importance, the remaining items can be combined for consideration. Since the major aerodynamic forces are the result of various pressures distributed on a surface, the *surface area* will be a major factor. *Dynamic pressure* of the airstream is another common denominator of aerodynamic forces and is a major factor since the *magnitude* of a pressure distribution depends on the source energy of the free stream. The remaining major factor is the relative *pressure distribution* existing on the surface. Of course, the velocity distribution, and resulting pressure distribution, is determined by the shape or profile of the surface and the angle of attack. Thus, any aerodynamic force can be represented as the product of three major factors:

the surface area of the object
the dynamic pressure of the airstream
the coefficient or index of force determined by the relative pressure distribution

This relationship is expressed by the following equation:

$$F = C_F q S$$

where

F = aerodynamic force, lbs.
C_F = coefficient of aerodynamic force
q = dynamic pressure, psf
$= \frac{1}{2}\rho V^2$
S = surface area, sq. ft.

In order to fully appreciate the importance of the aerodynamic force coefficient, C_F, the above equation is rearranged to alternate forms:

$$C_F = \frac{F}{qS}$$

$$C_F = \frac{F/S}{q}$$

In this form, the aerodynamic force coefficient is appreciated as the aerodynamic force per surface area and dynamic pressure. In other words, the force coefficient is a dimensionless ratio between the average aerodynamic pressure (aerodynamic force per area) and the airstream dynamic pressure. All the aerodynamic forces of lift and drag are studied on this basis—the common denominator in each case being *surface area and dynamic pressure.* By such a definition, a "lift coefficient" would be the ratio between lift pressure and dynamic pressure; a "drag coefficient" would be the ratio between drag pressure and dynamic pressure. The use of the coefficient form of an aerodynamic force is necessary since the force coefficient is:

(1) *An index of the aerodynamic force* independent of area, density, and velocity. It is derived from the relative pressure and velocity distribution.

(2) Influenced only by the shape of the surface and angle of attack since these factors determine the pressure distribution.

(3) An index which allows evaluation of the effects of compressibility and viscosity. Since the effects of area, density, and velocity are obviated by the coefficient form, compressibility and viscosity effects can be separated for study.

THE BASIC LIFT EQUATION. Lift has been defined as the net force developed perpendicular to the relative wind. The aerodynamic force of lift on an airplane results from the generation of a pressure distribution on the wing. This lift force is described by the following equation:

$$L = C_L q S$$

where

L = lift, lbs.
C_L = lift coefficient.
q = dynamic pressure, psf
$= \frac{1}{2}\rho V^2$
S = wing surface area, sq. ft.

The lift coefficient used in this equation is the ratio of the lift pressure and dynamic pressure and is a function of the shape of the wing and angle of attack. If the lift coefficient of a *conventional airplane wing planform* were plotted versus angle of attack, the result would be typical of the graph of figure 1.11. Since the effects of speed, density, area, weight, altitude, etc., are eliminated by the coefficient form, an indication of the true lift capability is obtained. Each angle of attack produces a particular lift coefficient since the angle of attack *is the controlling factor* in the pressure distribution. Lift coefficient increases with angle of attack up to the maximum lift coefficient, $C_{L_{max}}$, and, as angle of attack is increased beyond the maximum lift angle, the airflow is unable to adhere to the upper surface. The airflow then separates from the upper surface and stall occurs.

INTERPRETATION OF THE LIFT EQUATION. Several important relationships are

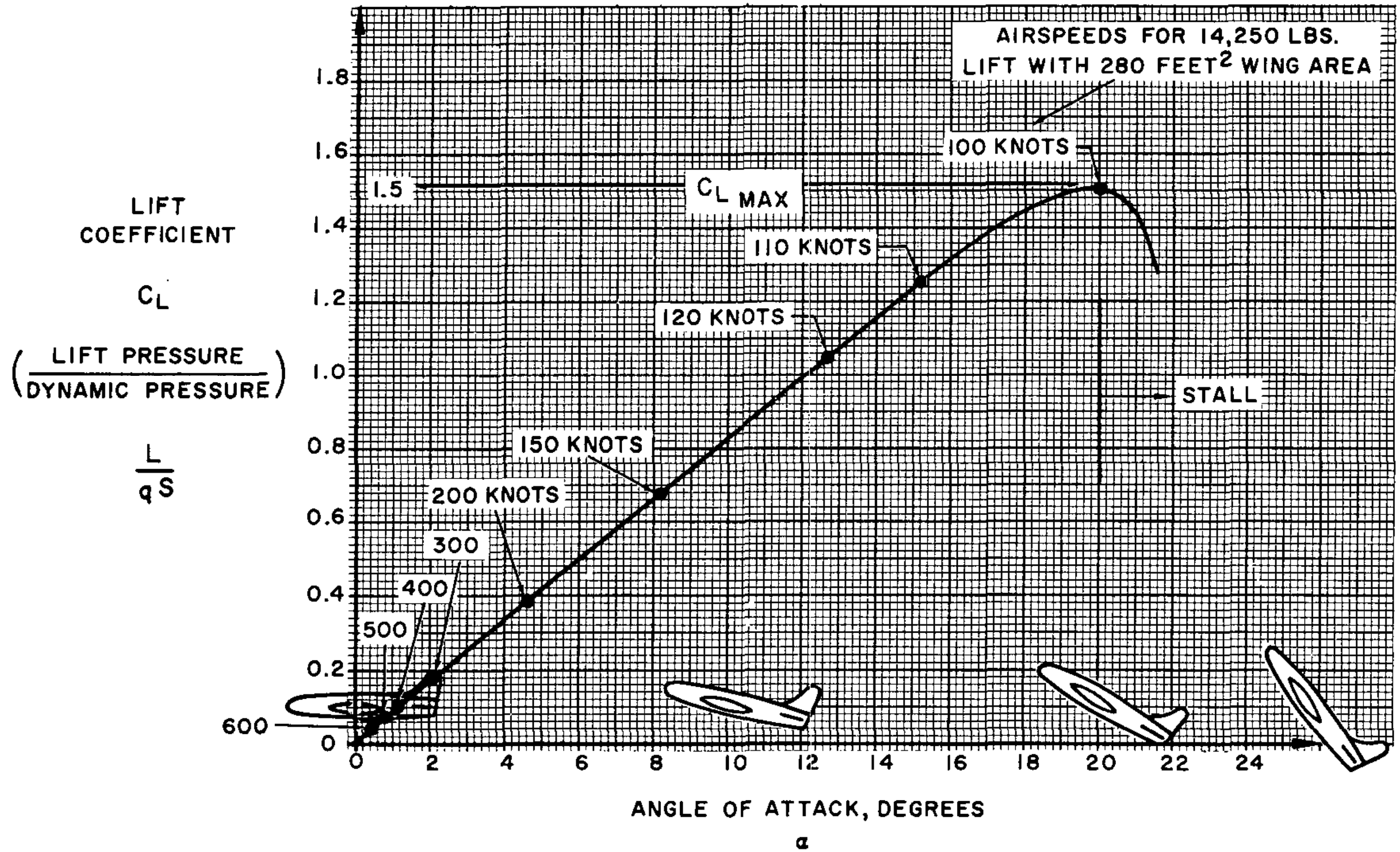


Figure 1.11. Typical Lift Characteristics

derived from study of the basic lift equation and the typical wing lift curve. One important fact to be appreciated is that the airplane shown in figure 1.11 stalls at the same angle of attack regardless of weight, dynamic pressure, bank angle, etc. *Of course, the stall speed* of the aircraft will be affected by weight, bank angle, and other factors since the product of dynamic pressure, wing area, and lift coefficient must produce the required lift. A rearrangement of the basic lift equation defines this relationship.

$$L=C_L qS$$

using $q=\frac{\sigma V^2}{295}$ (V in knots, TAS)

$$L=C_L\frac{\sigma V^2}{295}S$$

solving for V,

$$V=17.2\sqrt{\frac{L}{C_L\sigma S}}$$

Since the stall speed is the minimum flying speed necessary to sustain flight, the lift coefficient must be the maximum ($C_{L_{max}}$).

Suppose that the airplane shown in figure 1.11 has the following properties:

$$\text{Weight}=14{,}250\ \text{lbs.}$$

$$\text{Wing area}=280\ \text{sq. ft.}$$

$$C_{L_{max}}=1.5$$

If the airplane is flown in steady, level flight at sea level with lift equal to weight the stall speed would be:

$$V_s=17.2\sqrt{\frac{W}{C_{L_{max}}\sigma S}}$$

where

V_s = stall speed, knots TAS

W = weight, lbs. (lift = weight)

$$V_s=17.2\sqrt{\frac{(14{,}250)}{(1.5)(1.000)(280)}}$$

$$\doteq 100\ \text{knots}$$

Thus, a sea level airspeed (or *EAS*) of 100 knots would provide the dynamic pressure necessary at maximum lift to produce 14,250 lbs. of lift. If the airplane were operated at a higher weight, a higher dynamic pressure *would be required to furnish the greater lift* and a higher stall speed would result. If the airplane were placed in a steep turn, the greater lift required in the turn would increase the stall speed. If the airplane were flown at a higher density altitude the *TAS* at stall would increase. However, one factor common to each of these conditions is that the angle of *attack at $C_{L_{max}}$ is the same. It is important to* realize that stall warning devices must sense angle of attack (α) or pressure distribution (related to C_L).

Another important fact related by the basic lift equation and lift curve is variation of angle of attack and lift coefficient with airspeed. Suppose that the example airplane is flown in steady, wing level flight at various airspeeds with lift equal to the weight. It is obvious that an increase in airspeed above the stall speed will require a corresponding decrease in lift coefficient and angle of attack to maintain steady, lift-equal-weight flight. The exact relationship of lift coefficient and airspeed is evolved from the basic lift equation assuming constant lift (equal to weight) and equivalent airspeeds.

$$\frac{C_L}{C_{L_{max}}}=\left(\frac{V_s}{V}\right)^2$$

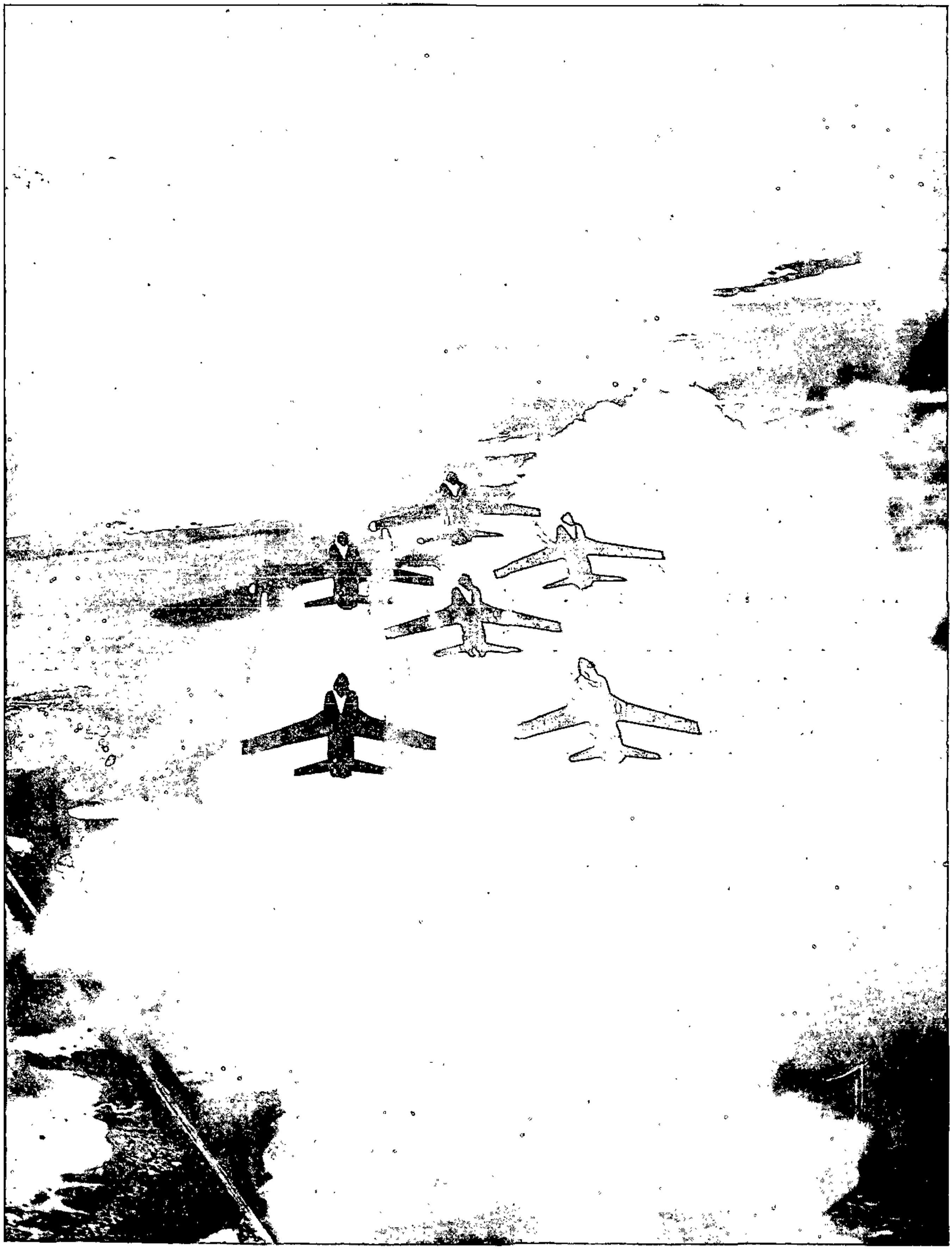
The example airplane was specified to have:

$$\text{Weight}=14{,}250\ \text{lbs.}$$

$$C_{L_{max}}=1.5$$

$$V_s=100\ \text{knots}\ EAS$$

The following table depicts the lift coefficients and angles of attack at various airspeeds in steady flight.



V, knots	$\frac{C_L}{C_{L_{max}}}=\left(\frac{V_s}{V}\right)^2$	C_L	α
100	1.000	1.50	20.0°
110	.826	1.24	15.2°
120	.694	1.04	12.7°
150	.444	.67	8.2°
200	.250	.38	4.6°
300	.111	.17	2.1°
400	.063	.09	1.1°
500	.040	.06	.7°
600	.028	.04	.5°

Note that for the conditions of steady flight, each airspeed requires a specific angle of attack and lift coefficient. This fact provides a fundamental concept of flying technique: *Angle of attack is the primary control of airspeed in steady flight.* Of course, the control stick or wheel allows the pilot to control the angle of attack and, thus, control the airspeed in steady flight. In the same sense, the throttle controls the output of the powerplant and allows the pilot to control rate of climb and descent at various airspeeds.

The real believers of these concepts are professional instrument pilots, LSO's, and glider pilots. The glider pilot (or flameout enthusiast) has no recourse but to control airspeed by angle of attack and accept whatever rate of descent is incurred at the various airspeeds. The LSO must become quite proficient at judging the flight path and angle of attack of the airplane in the pattern. The more complete visual reference field available to the LSO allows him to judge the angle of attack of the airplane more accurately than the pilot. When the airplane approaches the LSO, the precise judgment of airspeed is by the angle of attack rather than the rate of closure. If the LSO sees the airplane on the desired flight path but with too low an angle of attack, the airspeed is too high; if the angle of attack is too high, the airspeed is too low and the airplane is approaching the stall. The mirror landing system coupled with an angle of attack indicator is an obvious refinement. The mirror indicates the desired flight path and the angle of attack indicator allows precision control of the airspeed. The accomplished instrument pilot is the devotee of "attitude" flying technique—his creed being "attitude plus power equals performance." During a GCA approach, the professional instrument pilot controls airspeed with stick (angle of attack) and rate of descent with power adjustment.

Maneuvering flight and certain transient conditions of flight tend to complicate the relationship of angle of attack and airspeed. However, the majority of flight and, certainly, the most critical regime of flight (takeoff, approach, and landing), is conducted in essentially steady flight condition.

AIRFOIL LIFT CHARACTERISTICS. Airfoil section properties differ from wing or airplane properties because of the effect of the planform. Actually, the wing may have various airfoil sections from root to tip with taper, twist, sweepback and local flow components in a spanwise direction. The resulting aerodynamic properties of the wing are determined by the action of each section along the span and the three-dimensional flow. Airfoil section properties are derived from the basic shape or profile in two-dimensional flow and the force coefficients are given a notation of lower case letters. For example, a wing or airplane lift coefficient is C_L while an airfoil section lift coefficient is termed c_l. Also, wing angle of attack is α while section angle of attack is differentiated by the use of α_0. The study of section properties allows an objective consideration of the effects of camber, thickness, etc.

The lift characteristics of five illustrative airfoil sections are shown in figure 1.12. The section lift coefficient, c_l, is plotted versus section angle of attack, α_0, for five standard NACA airfoil profiles. One characteristic feature of all airfoil sections is that the slope of the various lift curves is essentially the same. At low lift coefficients, the section lift coefficient increases approximately 0.1 for each degree increase in angle of attack. For each of the airfoils shown, a 5° change in angle of

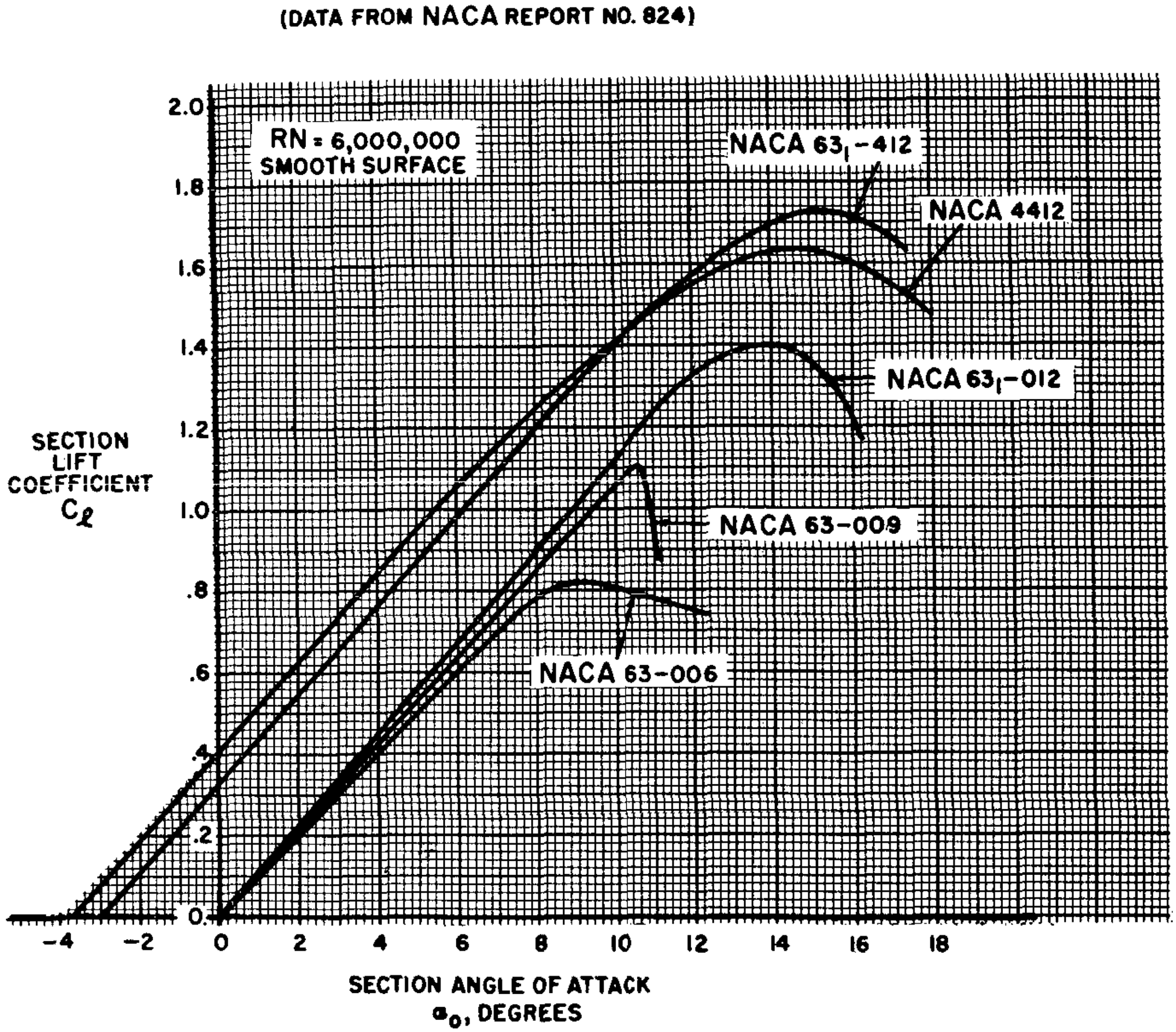


Figure 1.12. Lift Characteristics of Typical Airfoil Sections

attack would produce an approximate 0.5 change in lift coefficient. Evidently, lift curve slope is not a factor important in the selection of an airfoil.

An important lift property affected by the airfoil shape is the section maximum lift coefficient, $c_{l_{max}}$. The effect of airfoil shape on $c_{l_{max}}$ can be appreciated by comparison of the lift curves for the five airfoils of figure 1.12. The NACA airfoils 63–006, 63–009, and 63_1–012 are symmetrical sections of a basic thickness distribution but maximum thicknesses of 6, 9, and 12 percent respectively. The effect of thickness on $c_{l_{max}}$ is obvious from an inspection of these curves:

Section	$c_{l_{max}}$	α_0 for $c_{l_{max}}$
NACA 63–006	0.82	9.0°
NACA 63–009	1.10	10.5°
NACA 63_1–012	1.40	13.8°

The 12-percent section has a $c_{l_{max}}$ approximately 70 percent greater than the 6-percent thick section. In addition, the thicker airfoils have greater benefit from the use of various high lift devices.

The effect of camber is illustrated by the lift curves of the NACA 4412 and 63_1–412 sections. The NACA 4412 section is a 12 percent thick airfoil which has 4 percent maximum camber located at 40 percent of the chord. The NACA 63_1–412 airfoil has the same thickness and thickness distribution as the 63_1–012 but camber added to give a "design" lift coefficient (c_l for minimum section drag) of 0.4. The lift curves for these two airfoils show that camber has a beneficial effect on $c_{l_{max}}$.

Section	$c_{l_{max}}$	α_0 for $c_{l_{max}}$
NACA 63_1–012 (symmetrical)	1.40	13.8°
NACA 63_1–412 (cambered)	1.73	15.2°

An additional effect of camber is the change in zero lift angle. While the symmetrical sections have zero lift at zero angle of attack, the sections with positive camber have negative angles for zero lift.

The importance of maximum lift coefficient is obvious. If the maximum lift coefficient is high, the stall speed will be low. However, the high thickness and camber necessary for high section maximum lift coefficients may produce low critical Mach numbers and large twisting moments at high speed. In other words, a high maximum lift coefficient is just *one* of the many features desired of an airfoil section.

DRAG CHARACTERISTICS. Drag is the net aerodynamic force parallel to the relative wind and its source is the pressure distribution and skin friction on the surface. Large, thick bluff bodies in an airstream show a predominance of form drag due to the unbalanced pressure distribution. However, streamlined bodies with smooth contours show a predominance of drag due to skin friction. In a fashion similar to other aerodynamic forces, drag forces may be considered in the form of a coefficient which is independent of dynamic pressure and surface area. The basic drag equation is as follows:

$$D = C_D q S$$

where

D = drag, lbs.
C_D = drag coefficient
q = dynamic pressure, psf
$= \frac{\sigma V^2}{295}$ (V in knots, *TAS*)
S = wing surface area, sq. ft.

The force of drag is shown as the product of dynamic pressure, surface area, and drag coefficient, C_D. The drag coefficient in this equation is similar to any other aerodynamic force coefficient—it is the ratio of drag pressure to dynamic pressure. If the drag coefficient of a conventional airplane were plotted versus angle of attack, the result would be typical of the graph shown in figure 1.13. At low angles of attack the drag coefficient is low and small changes in angle of attack create only slight changes in drag coefficient. At

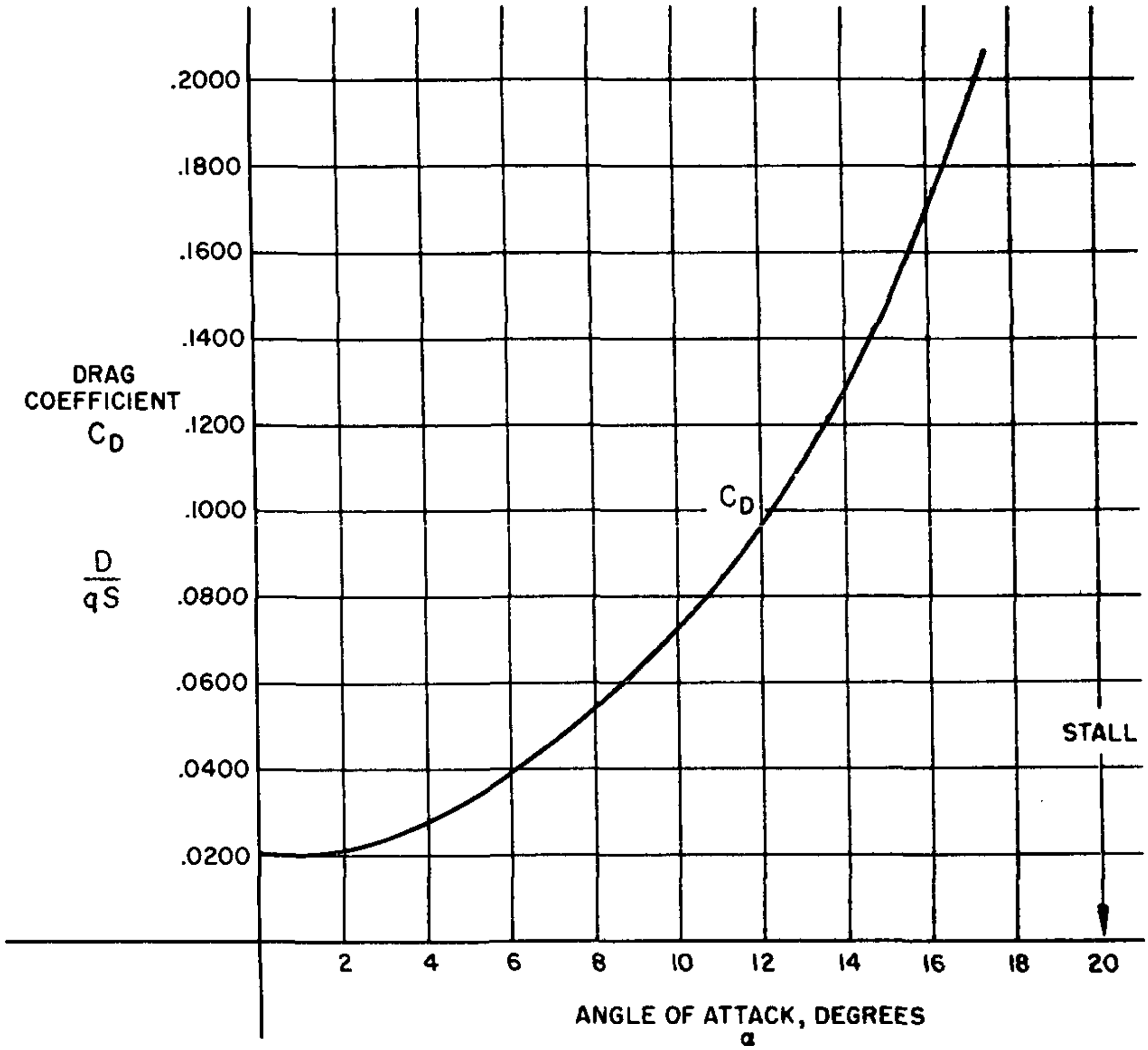


Figure 1.13. Drag Characteristics (sheet 1 of 2)

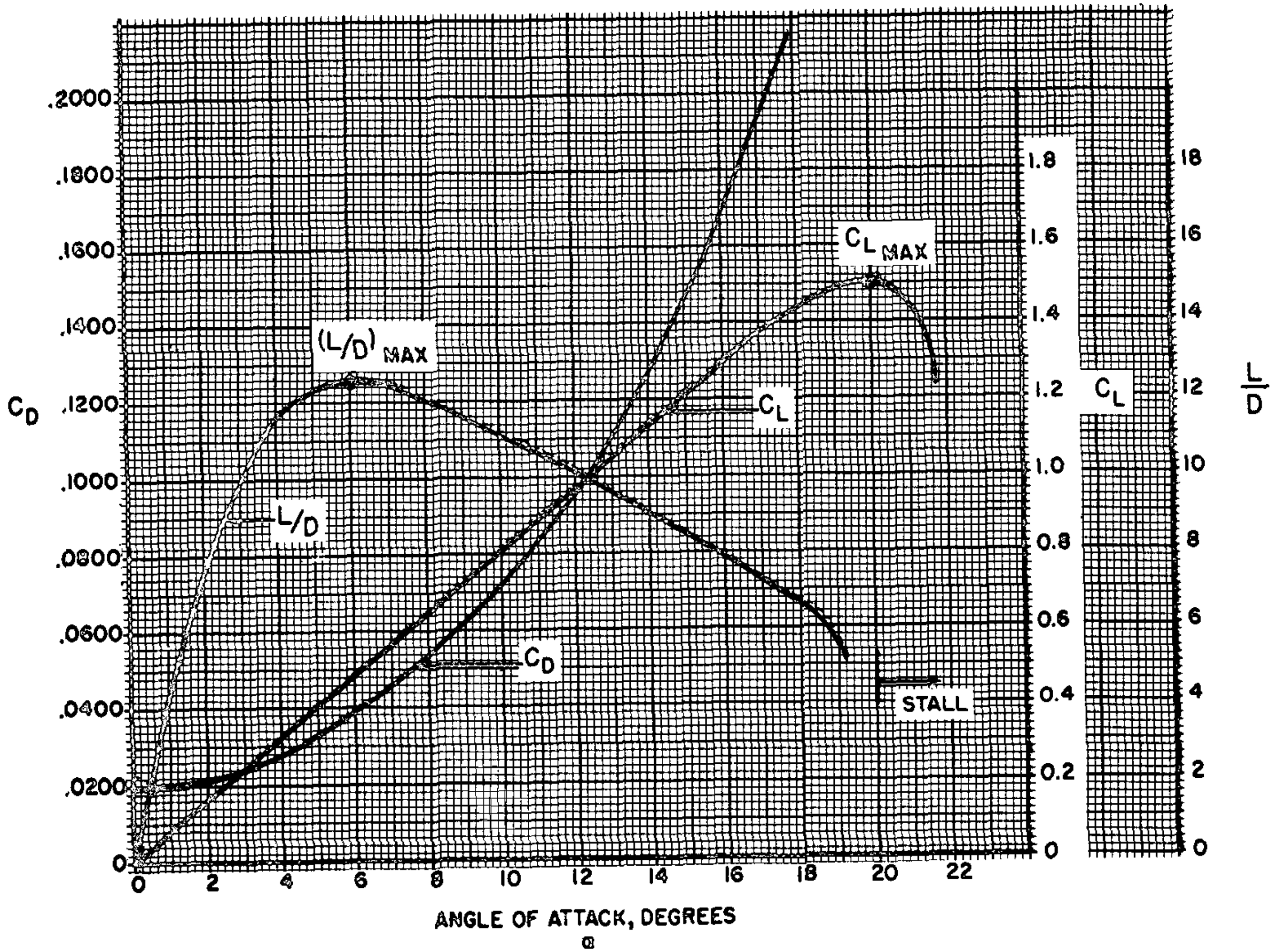


Figure 1.13. Drag Characteristics (sheet 2 of 2)

higher angles of attack the drag coefficient is much greater and small changes in angle of attack cause significant changes in drag. As stall occurs, a large increase in drag takes place.

A factor more important in airplane performance considerations is the lift-drag ratio, L/D. With the lift and drag data available for the airplane, the proportions of C_L and C_D can be calculated for each specific angle of attack. The resulting plot of lift-drag ratio with angle of attack shows that L/D increases to some maximum then decreases at the higher lift coefficients and angles of attack. Note that the maximum lift-drag ratio, $(L/D)_{max}$, occurs at one specific angle of attack and lift coefficient. If the airplane is operated in steady flight at $(L/D)_{max}$, the total drag is at a minimum. Any angle of attack lower or higher than that for $(L/D)_{max}$ reduces the lift-drag ratio and consequently increases the total drag for a given airplane lift.

The airplane depicted by the curves of Figure 1.13 has a maximum lift-drag ratio of 12.5 at an angle of attack of 6°. Suppose this airplane is operated in steady flight at a gross weight of 12,500 lbs. If flown at the airspeed and angle of attack corresponding to $(L/D)_{max}$, the drag would be 1,000 lbs. Any higher or lower airspeed would produce a drag greater than 1,000 lbs. Of course, this same airplane could be operated at higher or lower gross weights and the same maximum lift-drag ratio of 12.5 could be obtained at the same angle of attack of 6°. However, a change in gross weight would require a change in airspeed to support the new weight at the same lift coefficient and angle of attack.

Type airplane:	$(L/D)_{max}$
High performance sailplane	25–40
Typical patrol or transport	12–20
High performance bomber	20–25
Propeller powered trainer	10–15
Jet trainer	14–16
Transonic fighter or attack	10–13
Supersonic fighter or attack	4–9 (subsonic)

The configuration of an airplane has a great effect on the lift-drag ratio. Typical values of $(L/D)_{max}$ are listed for various types of airplanes. While the high performance sailplane may have extremely high lift-drag ratios, such an aircraft has no real economic or tactical purpose. The supersonic fighter may have seemingly low lift-drag ratios in subsonic flight but the airplane configurations required for supersonic flight (and high $[L/D]$'s at high Mach numbers) precipitate this situation.

Many important items of airplane performance are obtained in flight at $(L/D)_{max}$. Typical performance conditions which occur at $(L/D)_{max}$ are:

- maximum endurance of jet powered airplanes
- maximum range of propeller driven airplanes
- maximum climb *angle* for jet powered airplanes
- maximum power-off glide range, jet or prop

The most immediately interesting of these items is the power-off glide range of an airplane. By examining the forces acting on an airplane during a glide, it can be shown that the glide ratio is numerically equal to the lift-drag ratio. For example, if the airplane in a glide has an (L/D) of 15, each mile of altitude is traded for 15 miles of horizontal distance. Such a fact implies that the airplane should be flown at $(L/D)_{max}$ to obtain the greatest glide distance.

An unbelievable feature of gliding performance is the effect of airplane gross weight. Since the maximum lift-drag ratio of a given airplane is an intrinsic property of the aerodynamic configuration, gross weight will not affect the gliding performance. If a typical jet trainer has an $(L/D)_{max}$ of 15, the aircraft can obtain a maximum of 15 miles horizontal distance for each mile of altitude. This would be true of this particular airplane at any gross weight if the airplane is flown at the angle of attack for $(L/D)_{max}$. Of course, the gross weight would affect the glide airspeed necessary for this particular angle of attack but the glide ratio would be unaffected.

AIRFOIL DRAG CHARACTERISTICS. The total drag of an airplane is composed of the drags of the individual components and the forces caused by interference between these components. The drag of an airplane configuration must include the various drags due to lift, form, friction, interference, leakage, etc. To appreciate the factors which affect the drag of an airplane configuration, it is most logical to consider the factors which affect the drag of airfoil sections. In order to allow an objective consideration of the effects of thickness, camber, etc., the properties of two-dimensional sections must be studied. Airfoil section properties are derived from the basic profile in two-dimensional flow and are provided the lower case shorthand notation to distinguish them from wing or airplane properties, e.g., wing or airplane drag coefficient is C_D while airfoil section drag coefficient is c_d.

The drag characteristics of three illustrative airfoil sections are shown in figure 1.14. The section drag coefficient, c_d, is plotted versus the section lift coefficient, c_l. The drag on the airfoil section is composed of pressure drag and skin friction. When the airfoil is at low lift coefficients, the drag due to skin friction predominates. The drag curve for a conventional airfoil tends to be quite shallow in this region since there is very little variation of skin friction with angle of attack. When the airfoil is at high lift coefficients, form or pressure drag predominates and the drag coefficient varies rapidly with lift coefficient. The NACA 0006 is a thin symmetrical profile which has a maximum thickness of 6 percent located at 30 percent of the chord. This section shows a typical variation of c_d and c_l.

The NACA 4412 section is a 12 percent thick airfoil with 4 percent maximum camber at 40 percent chord. When this section is compared with the NACA 0006 section the effect of camber can be appreciated. At low lift coefficients the thin, symmetrical section has much lower drag. However, at lift coefficients above 0.5 the thicker, cambered section has the lower drag. Thus, proper camber and thickness can improve the lift-drag ratio of the section.

The NACA 63_1–412 is a cambered 12 percent thick airfoil of the "laminar flow" type. This airfoil is shaped to produce a design lift coefficient of 0.4. Notice that the drag curve of this airfoil has distinct aberrations with very low drag coefficients near the lift coefficient of 0.4. This airfoil profile has its camber and thickness distributed to produce very low uniform velocity on the forward surface (minimum pressure point well aft) at this lift coefficient. The resulting pressure and velocity distribution enhance extensive laminar flow in the boundary layer and greatly reduce the skin friction drag. The benefit of the laminar flow is appreciated by comparing the minimum drag of this airfoil with an airfoil which has one-half the maximum thickness—the NACA 0006.

The choice of an airfoil section will depend on the consideration of many different factors. While the $c_{l_{max}}$ of the section is an important quality, a more appropriate factor for consideration is the maximum lift coefficient of the section when various high lift devices are applied. Trailing edge flaps and leading edge high lift devices are applied to increase the $c_{l_{max}}$ for low speed performance. Thus, an appropriate factor for comparison is the ratio of section drag coefficient to section maximum lift coefficient with flaps—$c_d/c_{l_{m_f}}$. When this quantity is corrected for compressibility, a preliminary selection of an airfoil section is possible. The airfoil having the lowest value of $c_d/c_{l_{m_f}}$ at the design flight condition (endurance, range, high speed, etc.) will create the least section drag for a given design stall speed.

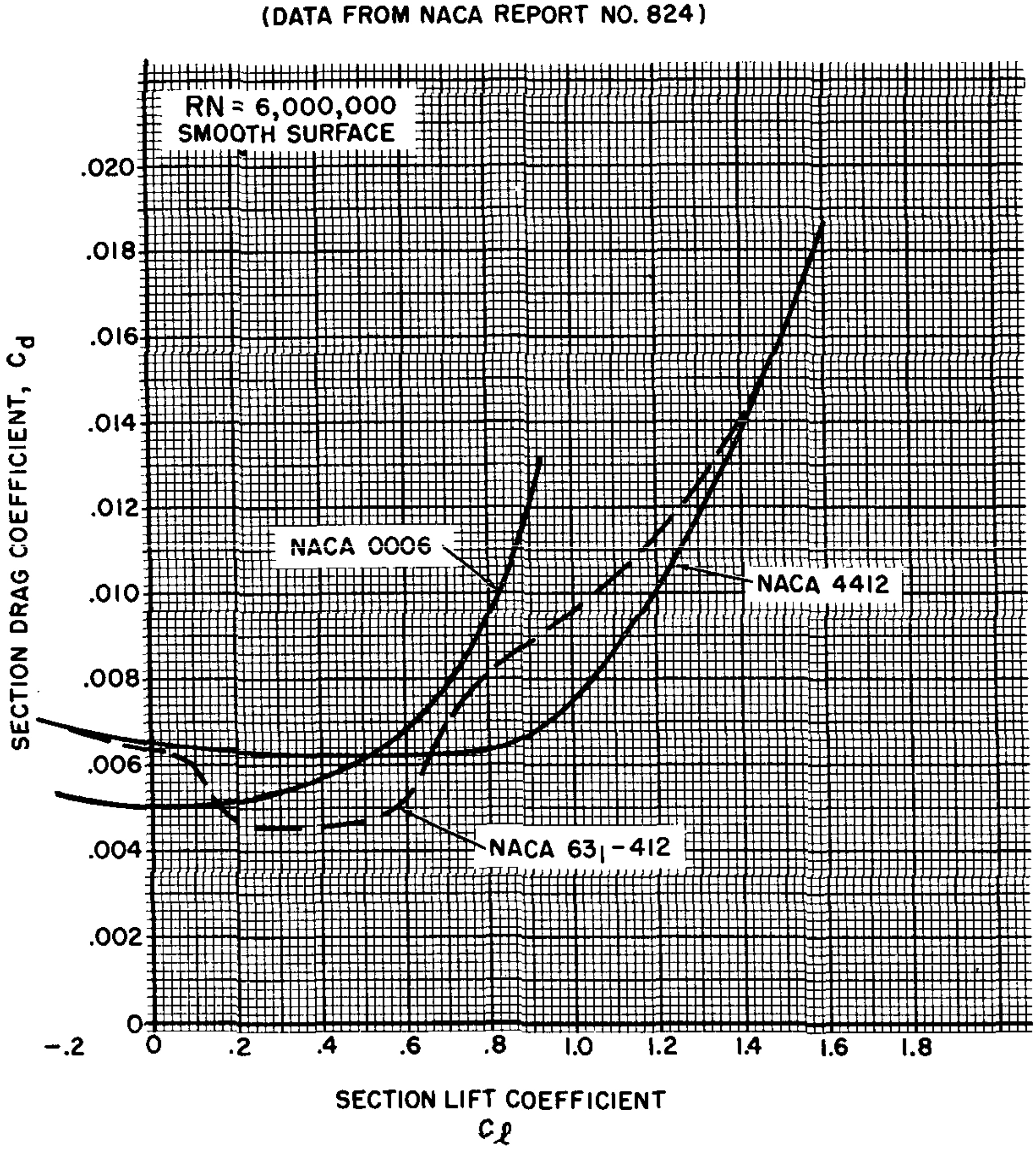


Figure 1.14. Drag Characteristics of Typical Airfoil Sections

FLIGHT AT HIGH LIFT CONDITIONS

It is frequently stated that the career Naval Aviator spends more than half his life "below a thousand feet and a hundred knots." Regardless of the implications of such a statement, the thought does connote the relationship of minimum flying speeds and carrier aviation. Only in Naval Aviation is there such importance assigned to precision control of the aircraft at high lift conditions. Safe operation in carrier aviation demands precision control of the airplane at high lift conditions.

The aerodynamic lift characteristics of an airplane are portrayed by the curve of lift coefficient versus angle of attack. Such a curve is illustrated in figure 1.15 for a specific airplane in the clean and flap down configurations. A given aerodynamic configuration experiences increases in lift coefficient with increases in angle of attack until the maximum lift coefficient is obtained. A further increase in angle of attack produces stall and the lift coefficient then decreases. Since the maximum lift coefficient corresponds to the minimum speed available in flight, it is an important point of reference. The stall speed of the aircraft in level flight is related by the equation:

$$V_s = 17.2\sqrt{\frac{W}{C_{L_{max}}\sigma S}}$$

where

V_s = stall speed, knots *TAS*
W = gross weight, lbs.
$C_{L_{max}}$ = airplane maximum lift coefficient
σ = altitude density ratio
S = wing area, sq. ft.

This equation illustrates the effect on stall speed of weight and wing area (or wing loading, W/S), maximum lift coefficient, and altitude. If the stall speed is desired in *EAS*, the density ratio will be that for sea level (σ= 1.000).

EFFECT OF WEIGHT. Modern configurations of airplanes are characterized by a large percent of the maximum gross weight being fuel. Hence, the gross weight and stall speed of the airplane can vary considerably throughout the flight. The effect of only weight on stall speed can be expressed by a modified form of the stall speed equation where density ratio, $C_{L_{max}}$, and wing area are held constant.

$$\frac{V_{s_2}}{V_{s_1}} = \sqrt{\frac{W_2}{W_1}}$$

where

V_{s_1} = stall speed corresponding to some gross weight, W_1
V_{s_2} = stall speed corresponding to a different gross weight, W_2

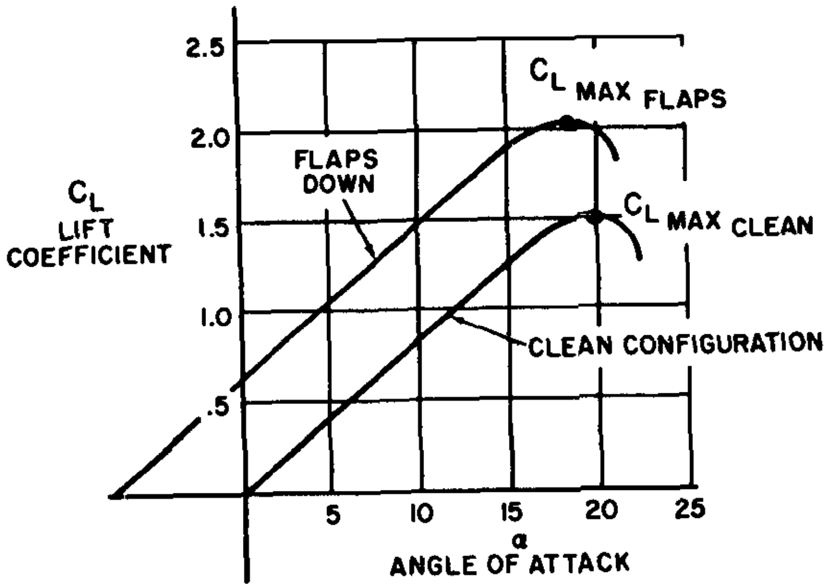
As an illustration of this equation, assume that a particular airplane has a stall speed of 100 knots at a gross weight of 10,000 lbs. The stall speeds of this same airplane at other gross weights would be:

Gross weight, lbs.	*Stall speed, knots EAS*
10,000	100
11,000	$100\times\sqrt{\frac{11,000}{10,000}}=105$
12,000	110
14,400	120
9,000	95
8,100	90

Figure 1.15 illustrates the effect of weight on stall speed on a percentage basis and will be valid for any airplane. Many specific conditions of flight are accomplished at certain fixed angles of attack and lift coefficients. The effect of weight on a percentage basis on the speeds for any specific lift coefficient and angle of attack is identical. Note that at small variations of weight, a rule of thumb may express the effect of weight on stall speed— "a 2 percent change in weight causes a 1 percent change in stall speed."

EFFECT OF MANEUVERING FLIGHT. Turning flight and maneuvers produce an effect on stall speed which is similar to the effect of weight. Inspection of the chart on figure 1.16 shows the forces acting on an airplane in a steady turn. Any steady turn requires that the *vertical* component of lift be equal to

EFFECT OF FLAPS



EFFECT OF WEIGHT ON STALL SPEED

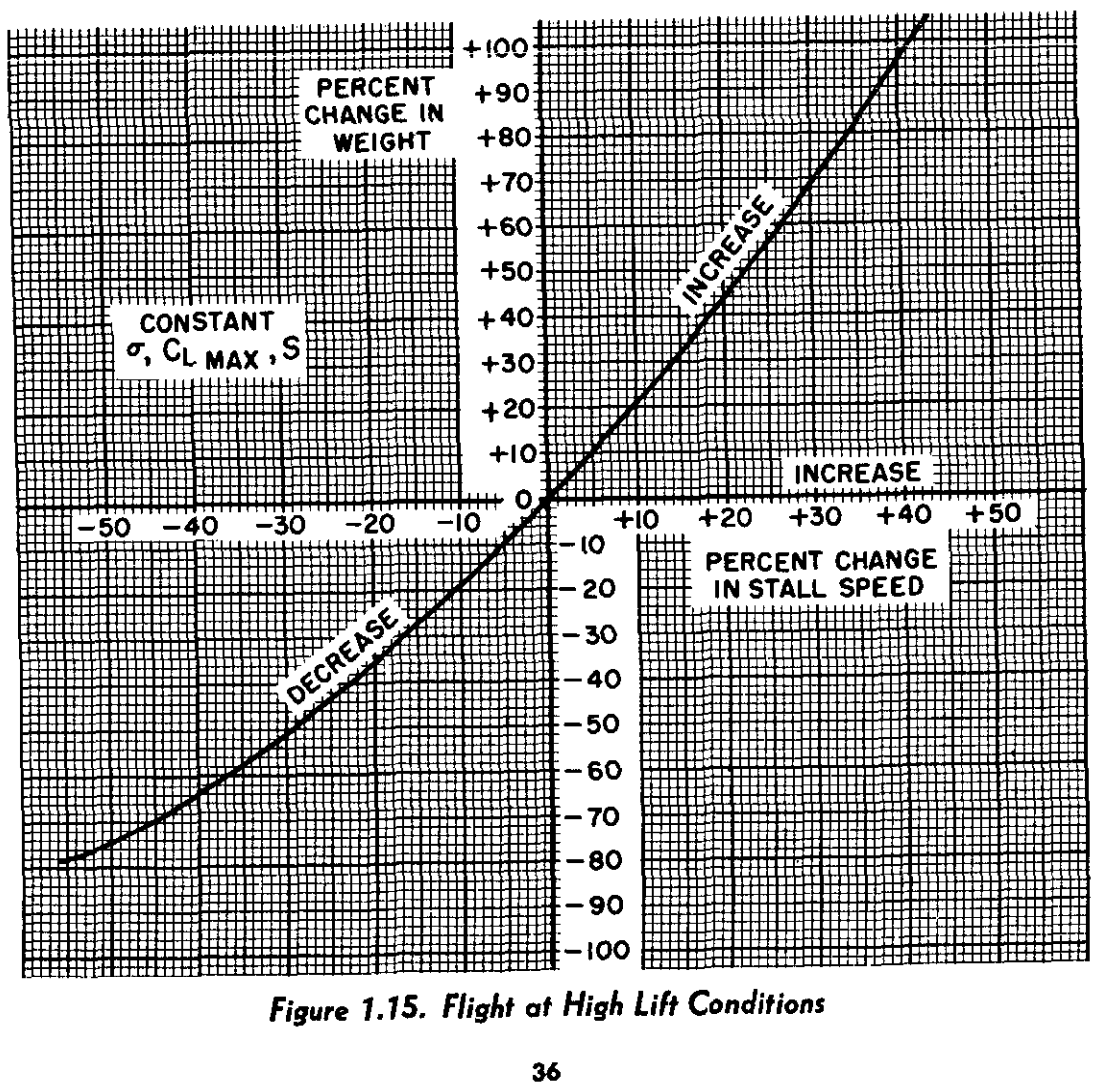


Figure 1.15. Flight at High Lift Conditions

weight of the airplane and the *horizontal* component of lift be equal to the centrifugal force. Thus, the aircraft in a steady turn develops a lift greater than weight and experiences increased stall speeds.

Trigonometric relationships allow determination of the effect of bank angle on stall speed and load factor. The load factor, n, is the proportion between lift and weight and is determined by:

$$n=\frac{L}{W}$$

$$n=\frac{1}{\cos\phi}$$

where

n = load factor (or "G")
$\cos\phi$ = cosine of the bank angle, ϕ (phi)

Typical values of load factor determined by this relationship are:

ϕ	0°	15°	30°	45°	60°	75.5°
n	1.00	1.035	1.154	1.414	2.000	4.000

The stall speed in a turn can be determined by:

$$V_{s\phi}=V_s\sqrt{n}$$

where

$V_{s\phi}$ = stall speed at some bank angle ϕ
V_s = stall speed for wing level, lift-equal-weight flight
n = load factor corresponding to the bank angle

The percent increase in stall speed in a turn is shown on figure 1.16. Since this chart is predicated on a steady turn and constant $C_{L_{max}}$, the figures are valid for any airplane. The chart shows that no appreciable change in load factor or stall speed occurs at bank angles less than 30°. Above 45° of bank the increase in load factor and stall speed is quite rapid. This fact emphasizes the need for avoiding steep turns at low airspeeds—a flight condition common to stall-spin accidents.

EFFECT OF HIGH LIFT DEVICES. The primary purpose of high lift devices (flaps, slots, slats, etc.) is to increase the $C_{L_{max}}$ of the airplane and reduce the stall speed. The takeoff and landing speeds are consequently reduced. The effect of a typical high lift device is shown by the airplane lift curves of figure 1.15 and is summarized here:

Configuration	$C_{L_{max}}$	α for $C_{L_{max}}$
Clean (flaps up)	1.5	20°
Flaps down	2.0	18.5°

The principal effect of the extension of flaps is to increase the $C_{L_{max}}$ and reduce the angle of attack for any given lift coefficient. The increase in $C_{L_{max}}$ afforded by flap deflection reduces the stall speed in a certain proportion, the effect described by the equation:

$$V_{s_f}=V_s\sqrt{\frac{C_{L_m}}{C_{L_{m_f}}}}$$

where

V_{s_f} = stall speed with flaps down

V_s = stall speed without flaps

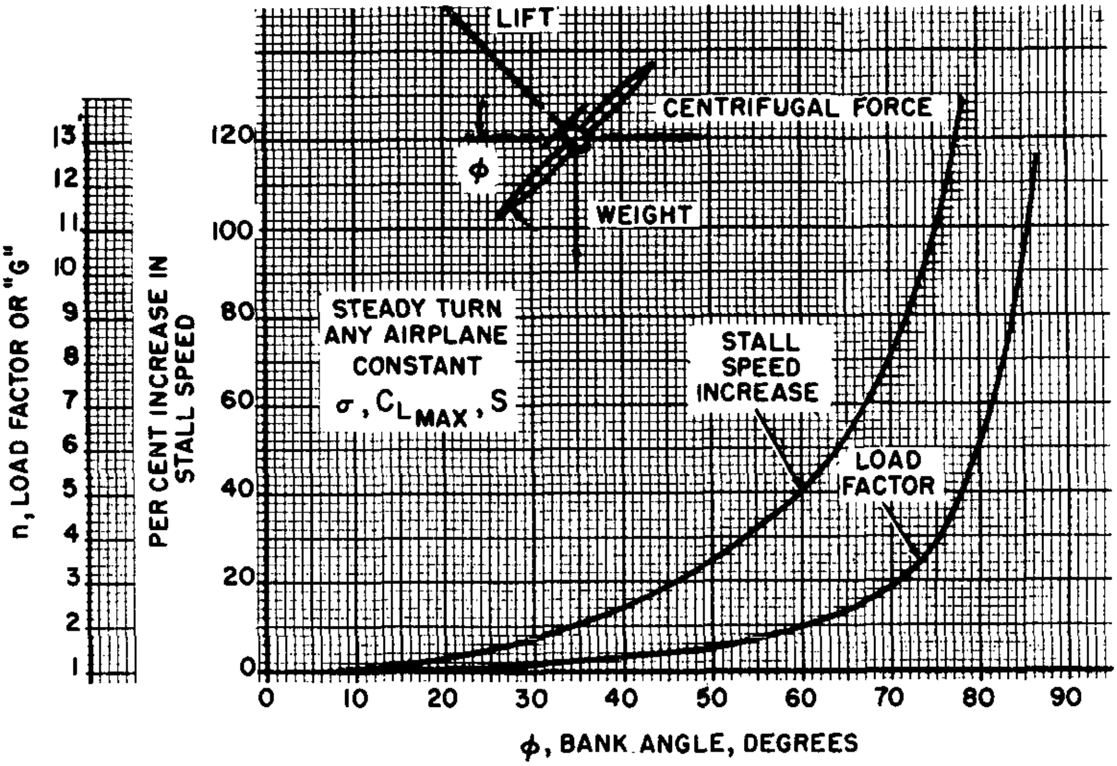
C_{L_m} = maximum lift coefficient of the clean configuration

$C_{L_{m_f}}$ = maximum lift coefficient with flaps down

For example, assume the airplane described by the lift curves of figure 1.15 has a stall speed of 100 knots at the landing weight in the clean configuration. If the flaps are lowered the reduced stall speed is reduced to:

$$V_{s_f}=100\times\sqrt{\frac{1.5}{2.0}}$$

$$=86.5\text{ knots}$$



EFFECT OF $C_{L_{MAX}}$ ON STALL SPEED

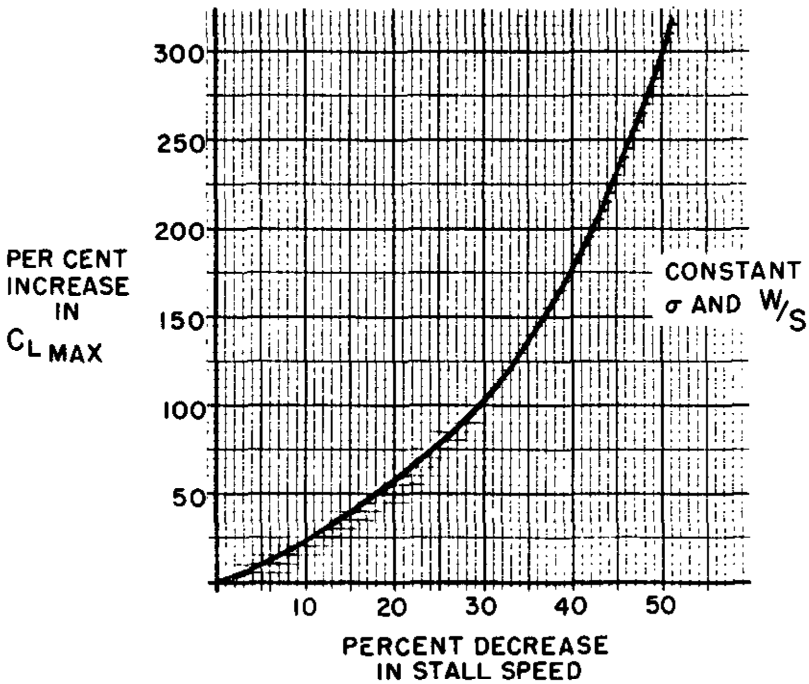


Figure 1.16. Flight at High Lift Conditions